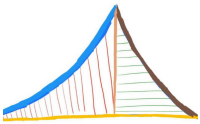
CBSE 2026
10th Class Maths
Solutions (65-3-1)



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About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9–12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share. License:
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First manual appeared in January 2018
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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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FORMULAE

1.1.1. Quadratic equations. For $ax^2 + bx + c = 0$,

$$\Delta = b^2 - 4ac, \qquad x = \frac{-b \pm \sqrt{\Delta}}{2a}, \qquad \alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}. \tag{1.1.1.1}$$

Equal real roots occur when $\Delta = 0$. For $y = ax^2+bx+c$, the vertex has x-coordinate $-b/(2a)$.

1.1.2. Tangents to a circle. A radius to the point of contact is perpendicular to the tangent. Tangents from the same external point are equal, and for radius r,

$$PT^2 = OP^2 - r^2. \tag{1.1.2.1}$$

1.1.3. Proportionality and similarity. If $AB \parallel EF$ in $\triangle DEF$, then

$$\frac{DA}{DE} = \frac{DB}{DF} = \frac{AB}{EF}, \qquad \frac{DA}{AE} = \frac{DB}{BF}. \tag{1.1.3.1}$$

For similar triangles, corresponding side ratios are equal and their area ratio is the square of that ratio.

1.1.4. Coordinate geometry. For $A(x_1, y_1)$ and $B(x_2, y_2)$,

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}, \qquad M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right). \tag{1.1.4.1}$$

If $AP : PB = m : n$, then $P = \left(\frac{nx_1+mx_2}{m+n}, \frac{ny_1+my_2}{m+n}\right)$.

1.1.5. Slope and parallel lines. The slope through two points is $m = (y_2 - y_1)/(x_2 - x_1)$. Non-vertical parallel lines have equal slopes.

1.1.6. Circle sectors and chords. For a central angle θ in degrees,

$$\text{arc length} = \frac{\theta}{360^\circ} 2\pi r, \qquad \text{sector area} = \frac{\theta}{360^\circ} \pi r^2. \tag{1.1.6.1}$$

A segment's area is the sector area minus the area of its triangle.

1.1.7. Right-triangle trigonometry. $\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$, $\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}$, and $\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$.

1.1.8. Mensuration.

$$V_{\text{cone}} = \frac{1}{3}\pi r^2 h, \qquad V_{\text{sphere}} = \frac{4}{3}\pi r^3, \qquad S_{\text{sphere}} = 4\pi r^2, \qquad S_{\text{cylinder,curved}} = 2\pi r h. \tag{1.1.8.1}$$

1.1.9. Probability. For equally likely outcomes, $P(E) = \frac{n(E)}{n(S)}$; $P(E^c) = 1 - P(E)$.

1.1.10. HCF and LCM. For positive integers a, b, $\text{HCF}(a, b)\,\text{LCM}(a, b) = ab$; the Euclidean algorithm finds the HCF by repeated division.

SOLVED QUESTIONS